COMPLETE HMI SPECIFICATION & REGISTER MAP

Target: Delta DOP-107 (Modbus TCP Master) | PLC: ABB AC500 (Slave @ 192.168.0.7 : 502)

CRITICAL ADDRESSING NOTE: ABB uses 0-based memory (%MW0). Delta uses 1-based Modbus (40001). All Delta Addresses in this document have already had the +1 offset applied (e.g., %MW1 = 40002).

1. Data Acquisition (DAQ) & Peak Telemetry (CRITICAL)

This section contains the high-speed data acquisition buffers and peak voltage captures generated during the test.

Variable Name	PLC Address	Delta Register	Data Type	Description / Formatting
DAQ Status / Log Complete	%MW100	**40101**	Word / Bit 0	Read Bit 0. Goes HIGH (1) when all 1,000 samples are fully populated.
Total Valid Samples	%MW100	**40101**	Word	Current number of samples in the buffer (0 to 1000).
Peak Voltage Phase UV	%MW31	**40032**	Word	Maximum Back-EMF UV during run. Format: 2 Fractional Digits (Value x100).
Peak Voltage Phase VW	**%MW32**	**40033**	**Word**	**Maximum Back-EMF VW. *Requires mapping to %MW32 in GVL_Modbus.**
Peak Voltage Phase WU	**%MW33**	**40034**	**Word**	**Maximum Back-EMF WU. *Requires mapping to %MW33 in GVL_Modbus.**

2. DAQ 1,000-Sample Trend Buffer (Memory Block)

Buffer Structure: The PLC logs an array of 1,000 structs (ST_TestSample). Each struct is exactly 5 Words long. The array starts at `%MW1000` (Delta `41001`) and ends at `%MW5999` (Delta `46000`). To plot this in the HMI, set your trend graphs to read interleaved memory starting at 41001 with an offset of 5.

Element (Sample 0)	PLC Addr	Delta Addr	Element (Sample 1)	PLC Addr	Delta Addr
Time (0.1s resolution)	%MW1000	**41001**	Time (0.1s resolution)	%MW1005	**41006**
Measured RPM	%MW1001	**41002**	Measured RPM	%MW1006	**41007**

Element (Sample 0)	PLC Addr	Delta Addr	Element (Sample 1)	PLC Addr	Delta Addr
Voltage UV (x100)	%MW1002	**41003**	Voltage UV (x100)	%MW1007	**41008**
Voltage VW (x100)	%MW1003	**41004**	Voltage VW (x100)	%MW1008	**41009**
Voltage WU (x100)	%MW1004	**41005**	Voltage WU (x100)	%MW1009	**41010**

3. Machine Setup & Parameters (HMI -> PLC)

Variable Name	PLC Address	Delta Register	Data Type	Limits / Behavior
Test Speed (RPM)	%MW1	**40002**	Word	Min: 0 | Max: 18000
Accel Time	%MW2	**40003**	Word	Seconds x10
Decel Time	%MW3	**40004**	Word	Seconds x10
HMI Heartbeat (Macro)	%MW4	**40005**	Word	Must increment every 1 sec to avoid Fault 11.
Test Duration	%MW5	**40006**	Word	Seconds
Fault Reset Cmd	%MW15	**40016**	Word / Bit 0	Momentary Pulse to clear Faults.

4. Machine State & VFD Telemetry (PLC -> HMI)

Variable Name	PLC Address	Delta Register	Data Type	Description
Machine State	%MW47	**40048**	Word	0=INIT, 1=IDLE, 2=WAIT_CLAMP, 3=WAIT_MOVE, 4=TEST_READY, 5=TESTING, 6=TEARDOWN, 7=WAIT_UNCLAMP, 8=FAULT
Fault Code	%MW48	**40049**	Word	1=E-Stop, 2=Air Loss, 3=Door, 4=ClampTmOut, 6=SledTmOut, 9=SensrConflict, 11=Watchdog, 14=VFDFault, 15=CommLoss
System Healthy	%MW49	**40050**	Word	1 = All interlocks safe. 0 = Safety Tripped.

Variable Name	PLC Address	Delta Register	Data Type	Description
HMI Comm Fault	%MW50	**40051**	Word	1 = Heartbeat timeout triggered.
Spindle Speed (Sensor)	%MW10	**40011**	Word	Live RPM read directly from the spindle proxy sensor.
VFD Actual Freq	%MW40	**40041**	Word	Fractional Digits: 2 (Value x 100)
VFD Actual RPM	%MW41	**40042**	Word	Feedback directly from VFD over Modbus.
VFD Actual Current	%MW42	**40043**	Word	Fractional Digits: 2 (Value x 100)